

Resolving the Sufficiency Deficit in Domain-Specific AI Guidance

A Theoretical Protocol for Calibrating and Validating Localized Small Language Models Under the Universal Core Framework (UCF)

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1. Abstract

Large language models are increasingly used for consumer-facing guidance in domains where errors can produce financial, legal, procedural, or physical harm. Prior work in the Universal Core Framework argues that these settings require evaluation standards beyond general benchmark performance: accuracy, completeness, actionability, safety, jurisdiction sensitivity, and transparency (Walcher 2026a). A companion Reference Implementation applies that framework across six applied consumer domains. It is designed to formalize the problem of sufficiency in high-stakes AI-generated guidance (Walcher 2026b). This paper is written in advance of that implementation. It assumes, for purposes of scoping a corrective architecture, that the anticipated sufficiency deficit will be confirmed in at least some domains. It then asks a forward-looking question: if that result holds, what kind of system would be worth building next? This paper proposes a theoretical protocol for determining whether localized small language models can reduce that deficit in selected domains.

Part I argues that broad-spectrum LLMs may remain structurally insufficient for direct consumer deployment on certain tasks. Generalized conversational competence does not guarantee localized, current, procedurally sequenced, and risk-sensitive advice. Part I also introduces a Multi-Domain SLM Feasibility Matrix. The matrix screens candidate domains by ground-truth availability, hyper-localization variance, and temporal volatility, and it is used to select insurance navigation as the preferred pilot. Part II specifies the protocol itself. It defines a deterministic retrieval-augmented generation scaffold with an authoritative source registry, an active-authority requirement, update-detection mechanics, and retrieval-failure governance. It defines a content map for corpus weighting, task mapping, edge-case codification, answer-pattern construction, and dataset splitting. It defines a build strategy that trains the SLM as a bounded domain-reasoning layer rather than a repository of current law. Finally, it defines a comparative validation protocol under the UCF, with an explicit statistical plan, that tests the resulting system against its base model, general-purpose LLMs, and retrieval-augmented alternatives.

The paper closes by distinguishing two possible deployment roles for a validated system: direct consumer guidance, and use as a domain-calibrated evaluator. It also considers explicitly what it would mean, and why it would still be useful, for this protocol to prove unnecessary.

The central contribution is methodological. Localized SLMs should not be presumed safer because they are smaller, open, or domain-specific. They should be treated as candidate corrective systems whose claims must be earned through feasibility analysis, source-governed retrieval, documented content engineering, controlled fine-tuning, and UCF-based validation. If general-purpose models turn out to be sufficient once properly evaluated, this protocol becomes unnecessary, and that would itself be a useful finding. If the anticipated deficit holds, this paper provides a disciplined path for adapting AI systems to high-stakes domains and for locating where domain-specific AI guidance fails.

Keywords: small language models; consumer guidance; AI evaluation; retrieval-augmented generation; domain-specific AI; insurance navigation; AI safety; jurisdiction sensitivity; model validation; Universal Core Framework

Part I. The Case for a Localized Corrective Architecture

Sections 2 and 3 make the argument: why an anticipated sufficiency deficit motivates considering a localized SLM at all, what would count as an appropriate candidate domain, and why insurance is the preferred pilot. A reader who accepts that case, or who is returning to consult the engineering protocol directly, can skip ahead to Part II.

2. Introduction: Rationale, Scope, and Status of This Paper

2.1 Status and Scope of This Paper

This paper is a prospective methodological document, not a build report. No localized SLM has been trained, and no comparative validation has been conducted. It is written before the companion Reference Implementation (Walcher 2026b) has collected data, and it does not assume the outcome of that study. Rather, it takes the anticipated sufficiency deficit as a working hypothesis worth planning for. If broad-spectrum LLMs are found insufficient for high-stakes consumer guidance in one or more evaluated domains, the field needs a disciplined answer to "what do we build instead." An ad hoc rush to fine-tune something domain-specific is not that answer. That is the gap this paper fills.

For the same reason, this paper does not name specific base models, fine-tuning libraries, embedding models, or serving stacks. The small-model and open-weights landscape is changing quickly. Any tool specified now would likely be obsolete, or superseded by a materially better option, by the time an implementation team acted on this protocol. Section 6 therefore specifies selection criteria rather than a fixed stack, so the protocol remains usable regardless of which specific tools are current when it is executed.

2.2 The Anticipated Sufficiency Deficit and the Case for a Smaller Unit of Control

The Universal Core Framework v1.0 was developed to evaluate AI-generated consumer guidance in domains where mistakes carry practical consequences, using six dimensions (Walcher 2026a). The companion Reference Implementation applies that framework across six applied domains. It is designed to test the concern that general-purpose LLMs may perform well on broad conversational tasks while remaining insufficient for high-stakes consumer guidance (Walcher 2026b). This paper proceeds on the assumption that concern will be borne out in at least some domains.

The anticipated sufficiency deficit refers to the gap between conversational adequacy and domain-specific deployability. A response may be fluent and superficially helpful while still being unsafe for direct consumer use. It may correctly define a concept but omit the procedural step that preserves a claim. It may describe a general rule but miss a jurisdictional exception. It may recommend an action before identifying the documents, deadlines, exclusions, or professional thresholds that govern the user's actual situation. Under the UCF, Tier 1 questions ask for definitions or static facts; Tier 2 requires ordered procedure; Tier 3 requires judgment under uncertainty (Walcher 2026a). A broad model may perform adequately on Tier 1, where the knowledge required is common and stable. Tier 2 and Tier 3 create a different burden: the model must sequence, qualify, localize, and recognize when the user's facts change the answer. The core problem is not hallucination alone but under-specification. Survey work on LLM hallucination has already cataloged the ways a model can state something false with fluent confidence (Ji et al. 2023; Huang et al. 2023), but the sufficiency deficit described here is a broader category. An

answer can be factually true and still insufficient, because it fails to tell the user when not to rely on it, when to preserve evidence, when to escalate, or when professional advice is necessary.

This paper proposes that localized small language models may offer a more tractable architecture for resolving that deficit in selected domains. The intuition is counterintuitive only if model size is treated as the primary proxy for guidance quality. Domain-specific consumer guidance is a constrained problem: the safest sufficient answer under known rules, sources, and boundaries, not an open-ended maximization problem. A smaller model can be more tightly bounded. It can be trained and evaluated within a narrower evidentiary environment, grounded in known statutes, regulations, standard forms, administrative guidance, and expert-authored edge cases. Its outputs can be constrained to answer patterns that prioritize caveats, jurisdiction checks, and escalation thresholds over conversational improvisation. The relevant comparison is therefore broad fluency versus calibrated sufficiency: bounded, not merely smaller.

The engineering case rests on three assumptions, examined in Section 3: that the target domain has accessible ground truth, manageable localization boundaries, and manageable temporal volatility. This paper does not argue that SLMs are a universal solution for consumer guidance. It argues only that they may be appropriate for domains with the right structure, and that this suitability must be demonstrated rather than assumed.

2.3 Preserving the UCF as the Metrology Standard, and Isolating Variables Through a Single-Domain Pilot

The proposed SLM protocol does not replace the Universal Core Framework; it depends on it. Without an external evaluation framework, a localized model could be declared successful merely because it was trained on the right corpus or sounded more cautious than a frontier LLM. Those are design features, not validation results. This paper therefore treats the UCF as fixed for purposes of validation. The six evaluation dimensions, the five-point scoring structure, the prompt complexity tiers, the independent-scoring requirement, and the descriptive/normative distinction all remain unchanged (Walcher 2026a). The SLM protocol is an intervention architecture; the UCF is the evaluation architecture. Keeping them separate keeps the comparison interpretable. If the same paper modified both at once, any improvement could be attributed either to the model or to the measurement system, and there would be no way to tell which.

A theoretical protocol for localized SLM development must also resist solving every consumer domain at once. Data ingestion, corpus weighting, statutory hierarchy, and edge-case drafting all depend on the structure of the selected domain. A multi-domain build at the first stage would introduce too many uncontrolled variables. The proposed protocol therefore begins with a single-domain pilot, chosen in Section 3, as a matter of methodological isolation. If the architecture fails in a domain with favorable structure, there is little reason to expect success elsewhere. If it succeeds, the architecture can be generalized cautiously to adjacent domains sharing the relevant feasibility characteristics.

2.4 From Evaluation Framework to Corrective Architecture

The first two papers in this series establish the measurement problem; this paper turns toward corrective architecture. If the UCF confirms a recurring sufficiency deficit in applied consumer guidance, the next research question is architectural: what kind of model system is most likely to reduce that deficit without creating new forms of opacity, overclaiming, or uncontrolled drift? A localized SLM is one candidate answer. It is not a replacement for all large models, nor a consumer-facing substitute for licensed professional judgment. It is a bounded, auditable, domain-specific guidance engine whose performance can be calibrated and validated against a stable external framework.

This paper does not claim that a localized SLM is the only, or even the most likely, corrective architecture. Other candidates are plausible and are not tested here. One is a frontier general-purpose LLM paired with a more rigorous, source-governed retrieval layer of the kind specified in Section 4, without any fine-tuning at all. Another is a constrained agentic workflow that decomposes a consumer question into verifiable sub-steps. A third is a hybrid symbolic-and-neural system that encodes source hierarchy as explicit rules rather than learned behavior. There may also be architectures not yet developed. The empirical comparison specified in Section 7 is designed precisely to test whether localization offers measurable gains beyond retrieval augmentation alone. If it does not, that comparison, not an a priori preference for smaller models, should settle the question. The present paper proposes one theoretically grounded candidate worthy of empirical evaluation, not a predetermined conclusion.

The remainder of Part I specifies the feasibility matrix used to select a candidate domain (Section 3). Part II then specifies the retrieval scaffold (Section 4), the data-ingestion and content-mapping process (Section 5), the SLM build and fine-tuning strategy (Section 6), the comparative validation protocol (Section 7), and the dual deployment role of a validated system (Section 8), before Section 9 considers the opposite outcome directly.

The claim advanced here is deliberately limited. Localized SLMs should not be presumed sufficient because they are specialized, any more than general-purpose LLMs should be presumed sufficient because they are larger or more fluent. The burden of proof remains on the model system in either direction. The contribution of this paper is to define how that burden can be met.

3. The Multi-Domain SLM Feasibility Matrix

3.1 Purpose

Not every consumer domain is an appropriate candidate for SLM development. Specialization alone does not make a model safe or maintainable. A domain-specific model can fail as severely as a general model if the underlying domain cannot be reduced to accessible ground truth, stable boundaries, and maintainable update cycles. This section asks a narrower question than the UCF's domain-eligibility criteria (Walcher 2026a, §7): assuming a domain is important enough to

evaluate, is it structurally suitable for a localized SLM build? A domain may be highly important for consumer protection while remaining poorly suited to compact model calibration. Health advocacy, for example, has abundant authoritative material, but its maintenance burden is driven by rapid change in clinical guidance and billing codes. Home remodeling poses real financial and physical risk, but it depends heavily on municipal codes and local enforcement practice that resist normalization.

The matrix evaluates candidate domains across three dimensions: ground-truth availability and accessibility, hyper-localization variance, and temporal volatility. These three are limited to the dimensions that determine whether a localized SLM can maintain sufficient guidance quality once built. Operational considerations such as cost, compute budget, licensing terms, deployment infrastructure, and organizational governance are deliberately excluded, because they affect implementation feasibility rather than epistemic feasibility. A domain can score well on all three matrix dimensions and still be impractical to build for reasons the matrix does not capture, and that distinction is intentional. The matrix screens for whether the architecture could work in principle, not whether a given team can afford to build it. It functions as a gatekeeping instrument, intended to prevent premature model building before a research team fine-tunes a model, constructs a retrieval corpus, or designs an evaluation suite.

3.2 Feasibility Dimension One: Ground-Truth Availability and Accessibility

A localized SLM depends on authoritative source material. If the domain lacks clean, accessible, hierarchically organized ground truth, calibration becomes unstable. The model may learn fragments of common practice, outdated statements, or inconsistent secondary summaries, without a reliable way to distinguish controlling authority from persuasive content. Ground truth does not require that every answer reduce to a single rule, but the model must have access to a dependable source hierarchy: primary authority, secondary interpretive material, standard forms, administrative guidance, and expert checklists.

Insurance illustrates a favorable case. Much of the domain is organized around policy language, state insurance codes, and regulator guidance, cleanly distinguishable from persuasive secondary material. Legal literacy has similarly strong source availability in many subdomains, but its statutes and case law are distributed across fifty-plus overlapping jurisdictions rather than a centralized statute-and-regulator structure. Existing work on domain-specific legal language models illustrates both the promise and the difficulty of this approach: systems such as ChatLaw pair a fine-tuned model with an external legal knowledge base to ground its answers (Cui et al. 2023), which is precisely the kind of source-fragmentation problem that makes Legal a harder candidate than Insurance for this protocol. Some consumer questions in the legal domain also approach individualized legal advice rather than pure legal information. Consumer fraud is more mixed, since durable consumer-protection principles must be separated from fast-moving scam typologies. Home remodeling illustrates the least favorable end: some ground truth exists in building codes and manufacturer instructions, but much of what consumers actually need

depends on property-specific conditions and informal trade knowledge that resists a clean hierarchy.

Ground-truth availability is rated high when authoritative materials are public, structured, stable, and organizable into a clear hierarchy. It is rated moderate when materials exist but are fragmented or mixed with rapidly changing interpretive content. It is rated low when the domain relies heavily on tacit knowledge, vendor-specific practice, or local custom.

3.3 Feasibility Dimension Two: Hyper-Localization Variance

Many consumer-guidance domains vary by geography, institution, policy, or factual setting. Variation itself does not defeat an SLM; the question is whether it can be bounded and encoded as explicit context variables. A domain is favorable when its localization boundaries are clean, so the model can identify the applicable jurisdiction, ask for missing location information, and retrieve the relevant authority before answering. A domain becomes less feasible when localization boundaries multiply into granular, undocumented local practice.

Insurance is favorable in this respect. Although there are many lines, carriers, and endorsements, the model can be scoped to a line of insurance and trained to treat state, policy type, and loss date as required context variables. Legal literacy is similar, since jurisdiction is usually expressible through state, federal, county, or procedural posture, though users may omit the necessary facts. Home buying sits in the middle, with national and state-level rules alongside county recording systems and title-company conventions that resist full normalization. Home remodeling is again the cautionary case, since even a correct general answer may fail under local permitting rules or municipal enforcement practice.

Hyper-localization variance is favorable when variation can be represented through a small number of explicit, retrievable variables. It is moderate when localization is important but partially bounded. It is unfavorable when local conditions are highly granular or too numerous to encode reliably.

3.4 Feasibility Dimension Three: Temporal Volatility and Maintenance Decay

A localized SLM is only useful if its domain knowledge can be maintained. A model that is accurate at release but decays rapidly may be more dangerous than a general model that openly retrieves current information. This is particularly true because consumers may not know whether an answer depends on current law or current administrative practice. Temporal volatility has two components: how fast the domain changes, and how detectable those changes are through known channels.

Insurance has a favorable temporal profile, because updates occur mainly through identifiable statutory, regulatory, and policy channels. Banking and financial guidance may also be feasible for stable regulatory guidance, though product terms shift more often; the finance sector already has a working precedent for large-scale domain-specific language modeling in BloombergGPT, a

model trained on a mix of general and proprietary financial data for finance-specific tasks (Wu et al. 2023), which is one reason Finance/Banking is included in Table 3.1 as a comparison point even though it falls outside the six domains evaluated in the companion Reference Implementation. Health advocacy is more difficult, since clinical guidance, coverage policies, and billing codes change frequently across many institutions. An SLM there may be feasible for procedural rights and appeal literacy, but not for current treatment guidance. Consumer fraud has a distinct volatility profile: legal principles are stable, but the threat environment changes quickly, so a static model trained on old scam patterns may fail precisely where consumers most need current guidance.

Temporal volatility is favorable when updates are predictable and tied to a small number of authoritative channels. It is moderate when durable principles exist but current applications change periodically. It is unfavorable when the domain decays quickly or accuracy depends on fast-moving facts distributed across many sources.

3.5 The Multi-Domain Feasibility Matrix

Table 3.1 applies the three dimensions to the six domains evaluated in the companion Reference Implementation (Walcher 2026b), plus Finance/Banking as an additional comparison vertical outside that six-domain scope. The ratings are theoretical classifications intended to guide SLM build selection, not empirical findings.

Consumer Vertical	Ground-Truth Availability & Accessibility	Hyper-Localization Variance	Temporal Volatility & Maintenance Decay	SLM Viability Score
Insurance	High: governed by explicit state statutes, administrative codes, and standard policy forms.	Moderate: variation is bounded strictly by state regulatory borders.	Low: changes occur along standard, manageable regulatory cycles.	HIGH (Preferred Pilot)
Legal	High but fragmented: codified via statutory text and case law, but distributed across 50+ state codes and overlapping case law rather than a centralized statute-and-regulator structure.	Moderate: rigid boundaries neatly isolated by state or federal jurisdiction.	Low: predictable legislative or judicial update cycles (annual/semi-annual).	HIGH
Finance / Banking (comparison only, outside the six-domain scope)	High: bounded by strict federal (e.g., SEC, CFPB) and state financial regulations.	Low: rules are largely uniform nationwide with minor state-level compliance deltas.	Moderate: shifting rate environments or consumer-protection acts require periodic updates.	HIGH
Consumer Fraud	Moderate: dependent on consumer-protection laws, state Attorney General bulletins, and FTC guidelines.	Moderate: bounded primarily by state-level jurisdiction and federal consumer rights.	High: fraud vectors and specific scams evolve rapidly, requiring nimble data ingestion.	MODERATE
Home Buying	Moderate: documented via	High: localized heavily by	Low: real estate transfer	MODERATE

Consumer Vertical	Ground-Truth Availability & Accessibility	Hyper-Localization Variance	Temporal Volatility & Maintenance Decay	SLM Viability Score
	federal acts (e.g., RESPA), state real estate disclosures, and lending guidelines.	county-level tax structures, municipal deed rules, and regional market norms.	frameworks and baseline laws remain relatively static over long periods.	
Health Advocacy	High: extensive availability of clinical guidance, CDC/FDA guidance, and federal health statutes.	Low: medical ground truth is largely national; administrative billing rules vary slightly by network.	Extreme: constant updates to drug approvals, federal insurance directives, and billing codes create rapid data decay.	MODERATE (High Maintenance Cost)
Home Remodeling	Low: fragmented across unwritten contractor norms, local trade practices, and unstructured vendor catalogs.	Extreme: bounded by hyper-local municipal zoning, building codes down to the ZIP code, and specific climate zones.	Low: local physical building codes change infrequently (decadal update cycles).	LOW (High Risk of Architectural Failure)

Table 3.1. Multi-Domain SLM Feasibility Matrix. Insurance and Legal both score in the favorable range; Insurance is selected as the pilot domain for the reasons given in Section 3.6. Finance/Banking is shown for comparison only and falls outside the six domains evaluated in the companion Reference Implementation (Walcher 2026b).

Figure 3.1 renders the same classifications as a heatmap. Color encodes favorability (green favorable, yellow moderate, red unfavorable) rather than a raw numeric score, since the three dimensions are not measured on a common quantitative scale. The heatmap is a visual index to Table 3.1, not an independent measurement.

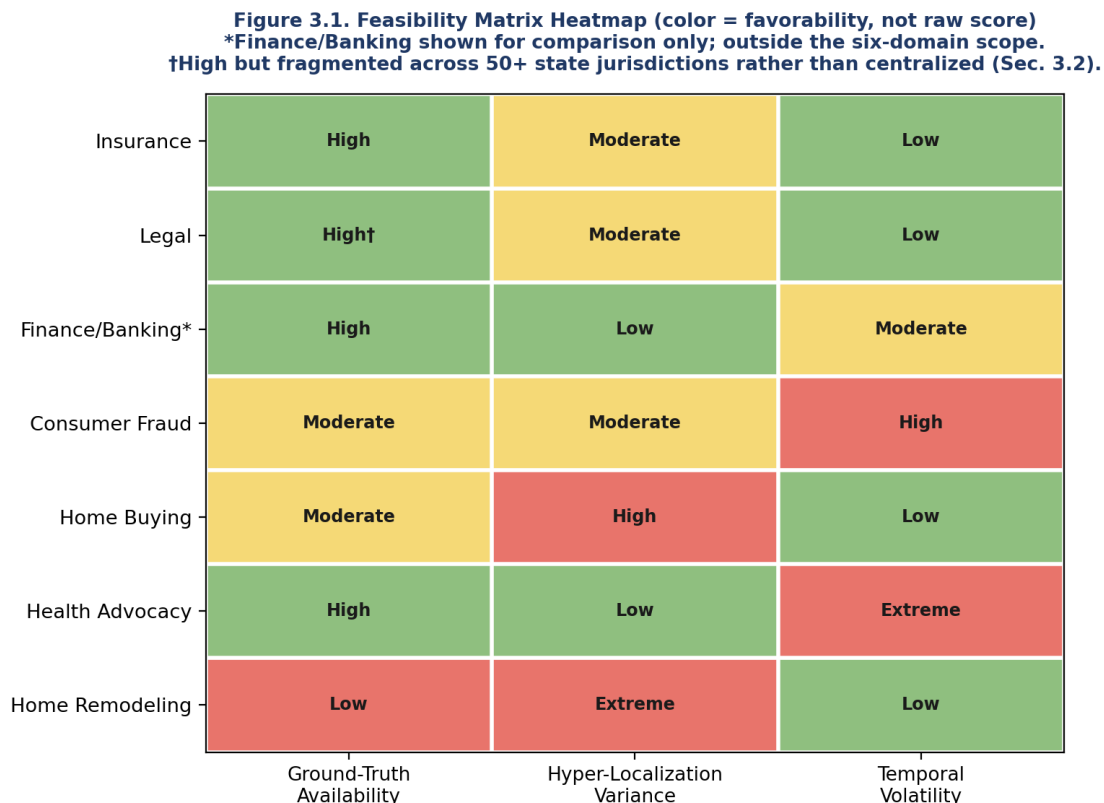


Figure 3.1. Feasibility Matrix Heatmap (color = favorability, not raw score). Finance/Banking is shown for comparison only and falls outside the six-domain scope.

3.6 Why Insurance Is the Preferred Pilot Domain

Insurance and Legal both score in the favorable range of the matrix. Insurance is selected as the pilot domain because its ground truth is more centrally codified: state statutes, administrative regulations, and standardized policy forms. Legal's ground truth is fragmented across overlapping case law and jurisdiction-specific procedure, and Legal carries an additional unauthorized-practice containment burden that Insurance does not. Insurance also offers a practical bridge to the evaluation architecture already described in the companion Reference Implementation (Walcher 2026b), so a localized insurance SLM can be evaluated against a prompt structure and scoring framework designed for the same domain.

A useful pilot domain should be hard enough that a general-purpose model is plausibly insufficient, and structured enough that a localized model is plausibly feasible. Insurance meets both conditions. A consumer asking how to respond to a denial letter, whether to provide a recorded statement, or how to escalate to a state insurance department requires the model to know what it does not know, ask for jurisdiction and policy language, and distinguish guidance from determination. These requirements map directly onto the UCF's six dimensions (Walcher 2026a).

This is a deliberately conservative framing, and it is worth stating plainly: a positive result in insurance would establish that the architecture is feasible in one structurally favorable domain, not that it is safe or feasible elsewhere. Insurance was chosen in part because a favorable outcome there implies the least about other domains. The pilot is hard enough to be informative but favorable enough that failure would be diagnostic rather than ambiguous. That is a narrower claim than choosing the domain most likely to impress a reader, and it is the more defensible one.

3.7 Domain Exclusion, Deferral, and Narrowing

The matrix also defines when a domain should be excluded, deferred, or narrowed rather than adopted wholesale. Home remodeling should not be selected for direct-guidance SLM development, given the risk of unsafe overgeneralization, though a narrower remodeling-adjacent system such as permit-checklist generation or contractor-interview preparation may remain feasible. Health advocacy should be narrowed rather than excluded. Patient-rights education and appeal-question preparation may be tractable where current coverage determinations and treatment recommendations are not. Consumer fraud should be treated as retrieval-dependent, since durable guidance about documentation and complaint channels is more stable than fast-moving scam typologies. Legal literacy should be treated as scope-sensitive: feasible for procedural education and issue-spotting only, with strong controls against individualized advice and unauthorized-practice boundaries. These rules keep the SLM solution a disciplined architecture rather than a general aspiration. A domain-specific model is only as credible as the domain-selection process that justified building it.

3.8 Implications for the Build Protocol

The feasibility matrix also shapes the build and validation burden that follows. A domain with high ground-truth availability should be expected to produce high accuracy and completeness if the model is well engineered, so a build that falls short in such a domain points to a corpus or engineering failure rather than an inherent limit of the approach. A domain with moderate localization variance, like insurance, should be evaluated closely on jurisdiction sensitivity, since that is where a favorable-but-not-perfect feasibility score is most likely to show strain. A domain with meaningful procedural risk should be evaluated closely on actionability and safety. A domain with temporal volatility should be tested through the maintenance and revalidation cycles specified in Section 7.6. Feasibility analysis is not merely a preliminary screen. It informs corpus design, source weighting, prompt construction, and validation emphasis throughout the rest of this protocol, and it helps distinguish, when a build underperforms, whether the likely cause is the build method, the corpus, the retrieval scaffolding, or the underlying feasibility assumption itself.

Part I has selected insurance as the pilot domain and specified why. Part II turns from domain selection to system stabilization. Even in a favorable domain, model weights alone are not enough. A localized SLM intended for high-stakes consumer guidance requires deterministic retrieval scaffolding, so that its reasoning remains tied to active, authoritative, version-controlled sources.

Part II. A Build and Validation Protocol

Sections 4 through 8 specify the deterministic retrieval scaffold, content-engineering process, fine-tuning strategy, comparative validation protocol, and deployment governance needed to build and test the system selected in Part I. This part presupposes the case made above: that insurance navigation is a feasible pilot domain for a localized SLM, worth testing against the Universal Core Framework.

4. The Deterministic RAG Scaffolding Protocol

4.1 Why Fine-Tuning Alone Is Insufficient

Selecting insurance as the pilot domain does not imply that model weights alone should carry the burden of consumer-guidance accuracy. Even in a favorable domain, a fine-tuned model is a static artifact between training cycles. Statutes are amended, administrative rules are revised, and policy forms are updated. If the model answers from stale internal representations, specialization creates misplaced confidence rather than improved sufficiency.

The proposed architecture therefore separates domain behavior from current authority. The SLM is responsible for disciplined reasoning within the domain: recognizing question type, identifying missing facts, applying source hierarchy, preserving caveats, and escalating when appropriate. The retrieval scaffold is responsible for supplying current, authoritative, jurisdiction-specific materials at response time. The general RAG literature describes combining parametric model knowledge with retrieved external information for knowledge-intensive generation (Lewis et al. 2020; Gao et al. 2023). This protocol narrows and hardens that concept for high-stakes consumer guidance. Retrieval does not merely enrich an answer; it constrains the answer to active authority.

The term deterministic reflects that the retrieval layer must draw from a controlled registry of approved sources, with source priority, jurisdiction labels, version metadata, and update monitoring. It is not an open-ended similarity search over arbitrary web content. For purposes of this protocol, deterministic retrieval means specifically that identical queries, submitted under identical registry conditions, retrieve the same governing authority according to explicit, disclosed priority rules. This does not mean the system is free of judgment. It means the judgment is rule-based and reproducible, rather than a probabilistic function of embedding similarity over an open web index. Recent surveys of RAG trustworthiness make a related point: retrieval improves faithfulness only when the system also governs what is allowed to be retrieved and how conflicts and gaps are handled, not merely how similarity is computed (Zhou et al. 2024).

4.2 The Authoritative Registry Blueprint

The first component of the scaffold is an Authoritative Registry: a structured directory of approved source locations, types, jurisdiction labels, update rules, and evidentiary weights. A suitable insurance registry distinguishes at least seven classes of sources:

- Primary statutory authority
- Administrative rules and regulations
- Official regulator guidance, such as consumer bulletins, claim-handling guides, and complaint instructions
- Standardized or approved policy materials, such as forms, endorsements, and exclusions
- Procedural materials, such as complaint forms, appeal instructions, and deadlines
- Expert-authored calibration materials, such as scenario maps and refusal thresholds

- Secondary explanatory materials, used only when clearly subordinate to primary and official sources

The registry encodes hierarchy directly rather than leaving it to the model's implicit judgment. A statute outranks a carrier FAQ. A current department bulletin outranks a blog post. A policy form controls over general explanation when the question turns on contract language. A state-specific source outranks generic national guidance when state law controls. For each entry, the registry should capture the following:

- Source ID, title, type, and jurisdiction
- Issuing authority and URL or storage location
- Effective date, last-retrieved date, and version identifier or revision date
- Update-frequency expectation and retrieval-priority weight
- Permitted-use category, known limitations, and supersession status

The permitted-use category matters in practice. A statute may support a controlling legal claim. A regulator consumer guide may support procedural explanation. A carrier FAQ may help identify industry practice, but it should never be treated as authoritative against policy language or state law. A secondary article may aid plain-language explanation, but it should not determine the answer.

The registry should also identify forbidden sources. For a high-stakes system, the retrieval layer excludes general web search results unless routed through review and formally added to the registry. This is one of the principal differences between deterministic RAG and ordinary open-web retrieval: the objective is controlled authority, not maximal coverage.

4.3 Source Ingestion and Normalization

Ingested materials must preserve the legal, procedural, and documentary structure that determines how a source should be interpreted. Ingestion should preserve source identity, since an unprovenanced chunk is ineligible for high-stakes generation. It should preserve structural location, since a retrieved sentence stripped of its exception can mislead. It should normalize jurisdiction metadata, assign source priority through retrieval weighting or reranking, detect supersession and staleness, and create citation-ready chunks so the audit layer can record which sources supported a given answer.

Dense passage retrieval and related embedding techniques support the semantic-matching component of this process (Karpukhin et al. 2020), but similarity alone is insufficient. A semantically similar passage from the wrong jurisdiction or a superseded source can be more dangerous than no retrieval at all. The scaffold therefore combines semantic retrieval with deterministic filters on jurisdiction, source type, effective date, and procedural stage.

4.4 Query Classification Before Retrieval

A consumer question should not reach the vector database before the system identifies the kind of guidance being requested. For insurance navigation, the classifier should identify at least the following attributes:

- Insurance line: auto, homeowners, renters, health, life, disability, long-term care, commercial, or unknown
- Question type: definition, coverage explanation, claim filing, denial response, appeal, complaint, bad-faith concern, documentation, settlement, or general planning
- Jurisdiction, policy dependency, and procedural posture (pre-claim, active claim, denial received, appeal pending, complaint filed, litigation threatened)
- Urgency: deadline-sensitive, claim-preservation issue, safety emergency, or ordinary informational request
- Professional threshold: whether the question approaches legal, licensed claim, medical, or tax advice

The classifier need not resolve the final answer. Its purpose is to prevent premature answering. If the state is missing and state law matters, the system asks for it or limits itself to caveated general guidance. Classification also scopes retrieval. A denial-letter question requires policy provisions, state claim-handling rules, and appeal deadlines. A definition question may require only a general source and a transparency caveat.

4.5 Retrieval, Reranking, and Context Assembly

Retrieval proceeds in stages. Deterministic filtering narrows candidates by jurisdiction, insurance line, source status, and permitted use. Semantic retrieval then identifies relevant passages within that filtered set. Source-priority reranking privileges primary and official sources over secondary explanation, and current sources over older ones. A context-completeness check determines whether the retrieved materials are sufficient to answer the question at all. If essential source types are missing, the model is instructed to say so rather than fill the gap.

Context assembly then orders the retrieved materials: controlling authority first, procedural authority second, policy-specific materials third, official consumer guidance fourth, expert calibration notes fifth, and secondary explanation last. This ordering ensures the model receives evidence in the same hierarchy the answer is expected to respect.

Formally, a composite ranking score for a candidate chunk can be expressed as a function of jurisdiction match, tier weight, and semantic similarity: $R = J_m \cdot (\alpha \cdot S_s + \beta \cdot W_t)$. Here J_m is a hard jurisdiction-match gate (0 or 1), S_s is semantic similarity, and W_t is the source's tier weight from the registry. S_s is typically a bounded similarity score, for example cosine similarity rescaled to [0,1]. W_t , by contrast, is drawn from a discrete five-tier classification (Tiers A through E). Because the two terms sit on different scales, both should be normalized to a shared [0,1] range before α and β are applied. Without this normalization step, a raw tier value could dominate, or be swamped by, the semantic-similarity term regardless of the weights chosen, and the ranking

would silently stop reflecting the intended priority structure. The exact functional form and the values of α and β should be fixed and disclosed as part of implementation documentation. This keeps the "deterministic" claim in this protocol's name backed by an auditable, reproducible procedure rather than an implicit heuristic.

The model is then instructed to answer only within the permitted scope defined by that assembly. If the context does not support a conclusion, the model does not infer one.

4.6 A Worked Example

The stages above are easiest to see applied to a single question. Consider a consumer who asks: "My homeowners insurer denied my claim for a burst pipe, and the adjuster said I have thirty days to appeal. Is that right?"

The classifier identifies insurance line as homeowners, question type as denial response, jurisdiction as unknown, policy dependency as high, procedural posture as denial received, and urgency as deadline-sensitive. Because jurisdiction is unknown, and the appeal deadline may be set by state law or policy language rather than a universal rule, deterministic filtering does not proceed to a state-specific answer. It either asks the user for their state or retrieves only materials explicitly labeled as general.

Suppose the user then supplies a state. Semantic retrieval surfaces candidate passages from that state's unfair-claims-settlement statute, the relevant department of insurance's appeal-procedure guidance, and a standard homeowners denial-letter template. Source-priority reranking places the statute and the department guidance ahead of a carrier FAQ that also matched the query. A context-completeness check notes that the user's own policy language and the specific denial letter have not been supplied, so the assembled context is sufficient to describe the general appeal process, but not to confirm the specific thirty-day figure the adjuster cited.

The model is therefore instructed to answer using the denial-response pattern from Section 5.3. It directs the user to the denial letter's stated reason and deadline. It explains that appeal windows can be set by either the policy or state procedure and that the two are not always identical. It recommends confirming the deadline in writing rather than relying on the adjuster's verbal statement, and it identifies the state department of insurance complaint pathway as a parallel option. It does not simply confirm "yes, thirty days is correct," because no retrieved source in this instance verified that figure against the user's actual policy or denial letter.

This is the Active Authority Requirement in operation. The system had enough grounding to describe the process correctly, but not enough to bless a specific number it could not verify, and it discloses that distinction rather than resolving it by assumption.

4.7 The Active Authority Requirement

The scaffold's core governance rule is the Active Authority Requirement: a model may not present localized, procedural, or rights-affecting guidance unless the answer is grounded in retrieved materials that are current, authoritative, and jurisdictionally appropriate. This yields four operational rules. There is no localized claim without jurisdictional grounding. There is no procedural claim without procedural authority. There is no policy-specific conclusion without policy language. There is no current-law claim from stale or unverified material. These rules convert retrieval from a helpful accessory into a condition of answer eligibility, binding answer confidence to source adequacy rather than to fluency.

4.8 Automated Update-Detection Mechanics

The registry pairs with an update-detection layer whose purpose is to identify change, trigger review, and prevent silent reliance on outdated content, not to interpret every change automatically. Detection should operate at several levels. A document-availability check confirms the source remains accessible. Hash comparison detects whether the document text has changed since the prior retrieval. Metadata comparison checks revision and effective dates. Structural comparison identifies changes in headings or procedural steps. Keyword alerting flags high-impact terms such as deadline, appeal, denial, cancellation, exclusion, revised, or emergency rule. Authority-status review detects whether a source has been archived, superseded, or withdrawn.

Detected changes trigger a graduated review workflow. Low-impact changes require corpus refresh and metadata update. Moderate changes require prompt-level regression tests. High-impact changes require expert review and revalidation of affected prompt categories. Every source should carry a version history of retrieval date, change status, review outcome, and validation action, so that a challenged answer can be traced to the exact source version available at the time.

4.9 Retrieval Failure Governance

Retrieval failure is an evidentiary failure, not a prompt-generation inconvenience, and it should not be hidden from the user. The protocol recognizes seven failure categories: no-source failure, wrong-jurisdiction failure, stale-source failure, conflict failure (retrieved authorities appear inconsistent and cannot be reconciled by hierarchy), insufficient-context failure, policy-dependency failure, and professional-threshold failure. Each category maps to a permitted response, such as a clarifying question, general education, identification of the missing document type, a regulator referral, or a professional-consultation recommendation, but never an invented answer covering the gap. This governance protects each UCF dimension directly: accuracy by refusing unsupported claims, completeness by naming missing source categories, actionability by offering a safe next step instead of a false conclusion, safety through escalation thresholds,

jurisdiction sensitivity through wrong-jurisdiction handling, and transparency through disclosed limitations (Walcher 2026a).

4.10 Auditability, Reproducibility, and the Relationship to Fine-Tuned Weights

Every response should produce a full audit record. This record should include:

- The user question and timestamp
- The model identifier and version
- The classifier output and any known or missing user facts
- The retrieval filters applied and the source IDs and versions retrieved
- The chunks passed to the model and their reranking scores
- The generated answer and any citations surfaced
- Any failure categories triggered or escalation logic applied

This record supports research reproducibility. It lets a reviewer determine whether a poor answer resulted from model reasoning, retrieval failure, source staleness, or registry design, a distinction essential to iterative improvement and to comparative validation in Section 7, where the analysis should record retrieval context alongside the final answer rather than the answer alone.

The scaffold does not eliminate the need for fine-tuning; it changes what fine-tuning is expected to accomplish. Fine-tuning should not be used primarily to memorize current law or policy details. That role belongs to the retrieval scaffold. Fine-tuning instead shapes domain behavior: classifying questions, respecting source hierarchy, identifying missing facts, preserving caveats, and escalating appropriately. This separation reduces maintenance burden, since a change in law or procedure can often be absorbed by updating the registry and corpus rather than retraining the model.

The resulting architecture has three layers: a behavioral layer, the fine-tuned SLM; an evidentiary layer, the deterministic RAG scaffold; and a validation layer, the UCF evaluation protocol. The SLM system is validated only if its observable outputs improve under the UCF dimensions, not because it has retrieval as a design feature.

5. Engineering the Data Ingestion Protocol: The Content Map

5.1 Purpose

The retrieval scaffold defines the authorized source environment; the Content Map converts that environment into a structured corpus supporting both retrieval and model training. Data selection determines model behavior in a high-stakes domain. A corpus overrepresenting general explanatory articles produces a fluent but weakly authoritative model. One overrepresenting statutes without procedural materials produces a legally dense but practically unhelpful model. One overrepresenting carrier materials risks reproducing institutional framing rather than consumer-protective guidance. One excluding edge cases performs adequately on clean

questions while failing on ambiguous or adversarial ones. The goal is therefore sufficient, weighted, auditable ingestion. The model should learn the structure of the domain, not merely its vocabulary.

The Content Map has five functions. It defines the ground-truth corpus. It assigns authority weights and permitted uses. It maps sources to consumer guidance tasks. It codifies edge cases and refusal thresholds. It produces training-ready and retrieval-ready artifacts that can later be validated under the UCF. This section uses insurance navigation as the reference domain. The same content-engineering logic would need to be adapted, not copied, for any other domain, since a universal content map would defeat the purpose of localized modeling.

Figure 5.1 places this content-engineering work inside the full pipeline specified across Sections 4 through 8, from authoritative sources through continuous update detection after deployment.

Figure 5.1. Content-Engineering and Deployment Pipeline

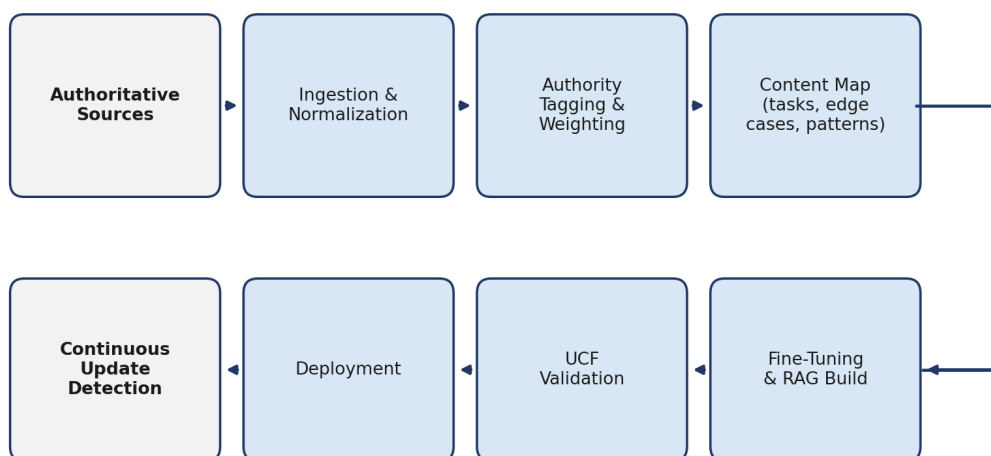


Figure 5.1. Content-Engineering and Deployment Pipeline.

5.2 Corpus Selection and Source Weighting

The corpus should be organized around six primary source classes:

- Statutory and regulatory authority
- Official regulator guidance, such as claim-filing, denial-dispute, and complaint procedures
- Policy and form materials, such as standard forms, endorsements, exclusions, and sample denial-letter language
- Procedural and documentation materials, such as checklists, appeal templates, and evidence-preservation guides

- Expert-authored calibration materials, such as scenario maps, safe-answer templates, and refusal triggers
- Secondary explanatory materials, subordinate to the first five

No source should be ingested unless the research team can answer what authority it has, what jurisdiction or policy context it applies to, what consumer task it supports, and what the risk is if the model overuses or misuses it. Sources that fail this test go to a secondary review queue rather than the active corpus.

Weighting should follow an explicit five-tier hierarchy. Tier A is controlling authority: statutes, regulations, and policy contract language. Tier B is official procedural guidance. Tier C is standardized policy and claim materials. Tier D is expert calibration material, which teaches the model how to behave when authoritative materials alone do not produce a safe answer. Tier E is secondary explanation, which may improve readability but never overrides higher tiers. This hierarchy should be implemented in both retrieval and training. Training examples should teach the model to acknowledge it explicitly, for example that coverage depends on policy language even when a regulator guide points toward a different practical answer. The weighting scheme should also define prohibited uses. A carrier FAQ is not a neutral statement of consumer rights. An educational article should not be used to state a state-specific deadline. A policy form from one state should not be generalized to another.

5.3 Task Mapping and the Answer Pattern Library

Consumers ask what something means, what to do next, whether a denial seems wrong, or when to get help, not for statutes in the abstract. The Content Map connects source types to these guidance functions. The protocol defines eight task categories for the insurance pilot: definition and concept explanation; coverage and exclusion literacy; claim filing and documentation; denial response and appeal; complaint and regulator escalation; bad-faith or unfair-claims concern; recorded statements and insurer communications; and deadline-sensitive questions. Each maps to its own primary source needs and required model behavior. A coverage question, for example, needs policy provisions and exclusions, and the model should refuse to determine coverage without policy language. A denial-response question needs the denial letter's structure and appeal procedures, and the model should direct the user to the stated reason and deadlines without concluding the denial was improper before review.

A localized SLM should also answer in repeatable, safe patterns rather than ad hoc structures. The Answer Pattern Library specifies the order and safeguards for each task category without predetermining content. A definition pattern defines the term, explains why it matters, identifies where to look in the policy, states that wording may change the result, and offers safe next steps. A procedure pattern identifies the goal, lists steps in order, identifies needed documents, flags deadlines or state dependence, and identifies escalation options. A denial-response pattern directs the user to the denial reason and cited provision, lists evidence to gather, identifies appeal and regulator options, and warns about deadlines and releases without concluding the denial was

wrongful. A coverage-literacy pattern explains that coverage depends on policy language and states what cannot be determined from the facts given. A bad-faith pattern notes jurisdiction-specificity and lists warning signs without a legal conclusion. A missing-facts pattern identifies the missing facts and why each matters before concluding. A refusal-or-escalation pattern states plainly that the request cannot be safely answered as asked, and recommends the appropriate professional, regulator, or document-review path.

These patterns should be embedded in supervised training examples and used as system-level constraints during generation. They also provide a basis for later scoring: a response following the appropriate pattern is more likely to score well on completeness, actionability, safety, jurisdiction sensitivity, and transparency.

5.4 Edge-Case Codification

Edge-case codification is the most important content-engineering step, because most unsafe consumer-guidance failures occur outside clean definitional questions. A model can appear competent when asked "What is a deductible?" and fail when asked, "My insurer denied my water damage claim, but the adjuster told me not to worry about the deadline. Should I wait?"

The edge-case library should include at least the following categories, each annotated with intended model behavior, required caveats, needed source types, and implicated UCF dimensions:

- Missing jurisdiction
- Missing policy language
- Deadline ambiguity
- Institutional pressure, such as pressure to sign a release, give a recorded statement, or withdraw a complaint
- Emotional distress
- Adversarial framing, such as a request for help exaggerating damages or withholding information
- Conflicting authorities
- Multi-party complexity
- Coverage-versus-procedure confusion
- Professional-boundary crossing

In each case the model's required behavior follows the same logic. Ask for what is missing. Flag risk rather than resolve it. Refuse and redirect where the request is adversarial. Identify hierarchy rather than adjudicate conflicting sources.

Structuring edge-case drafting this way turns it into a repeatable method rather than an editorial exercise. It also gives the eventual instruction dataset a documented library of adversarial and ambiguous scenarios. A synthetic-generation pipeline, for example a frontier model producing draft scenarios under domain-expert supervision with every draft manually verified before entering

the training set, could scale this library beyond what expert authors alone could produce in the same time.

5.5 Training Artifacts, Dataset Splits, and Quality Assurance

The Content Map should produce five distinct artifact types rather than one undifferentiated corpus. Retrieval chunks are source-preserving segments optimized for accurate selection and citation. Instruction examples are supervised examples pairing a question with a safe answer, required caveats, and prohibited conclusions. Contrastive examples pair an insufficient answer, plausible and friendly but missing a safeguard, with a corrected one. Edge-case scenarios cover the categories in Section 5.4. Evaluation prompts are held out from training and mapped to UCF dimensions, tiers, and expected failure modes.

The dataset should be split by scenario rather than by document, so training and test sets do not contain near-identical versions of the same situation. The split should preserve representation across complexity tiers and across failure modes, so the model is not evaluated only on ideal questions.

Before training begins, the corpus should pass six quality checks. An authority check confirms every source has an assigned authority level. A jurisdiction check confirms every jurisdiction-dependent source is labeled. A currency check confirms every source has a retrieval date and monitoring status. A conflict check confirms known conflicts are documented and resolved by hierarchy or flagged for expert review. A leakage check confirms evaluation prompts do not appear in training data. A behavioral check reviews training examples for overconfident conclusions or weak caveats, since the goal is calibrated sufficiency, not maximal caution.

All content-engineering decisions should be documented well enough for another research team to reconstruct the corpus logic even without access to every underlying source file. This supports reproducibility and lets a model failure be distinguished from a corpus-design failure. This documentation also enforces scope: a model trained for insurance navigation should be documented as not trained for legal representation, medical advice, or claim-adjusting services, consistent with the UCF's general restriction against overclaiming beyond the evidence generated by the study design (Walcher 2026a).

At this point the research team has defined the authorized corpus, source hierarchy, task categories, edge cases, answer patterns, and dataset splits, but has not yet fine-tuned a model. That sequencing is intentional. In high-stakes consumer guidance, training should not begin until the knowledge environment and desired answer behavior are specified. The next section turns to the model build itself.

6. SLM Build Strategy and Fine-Tuning Protocol

6.1 Purpose and Tool-Agnosticism

This section specifies how a localized SLM should be selected and trained once the retrieval scaffold and content map are in place. The build is not designed to produce a generally capable conversational model. Its success criterion is improved sufficiency on domain-specific guidance, especially on Tier 2 and Tier 3 tasks involving procedural sequencing, jurisdictional caution, and escalation.

Specific base models, adapter methods, quantization approaches, and serving stacks are deliberately not named. The open-weights and fine-tuning landscape moves quickly. A tool prescribed today would likely be superseded, or simply outdated, by the time an implementation team executed this protocol. Section 6.2 therefore specifies selection criteria that should remain stable across implementations even as the specific tools that satisfy them change.

The build has five objectives. It should preserve enough general language competence to communicate clearly with consumers. It should introduce domain-specific discipline without encouraging unsupported determinations. It should teach the model to follow the insurance source hierarchy. It should teach the model to recognize missing facts, retrieval gaps, and professional boundaries. It should maintain compatibility with deterministic RAG, audit logging, and UCF validation.

6.2 Base Model Selection Criteria

The target class is an open-weights, instruction-tuned model compact enough for controlled deployment while retaining sufficient instruction-following capacity for consumer-facing guidance. The exact parameter threshold should be declared at implementation time rather than fixed here.

Selection should be justified against seven criteria:

- Open-weights availability, permitting local deployment, fine-tuning, and reproducible research use
- Instruction-following competence
- Adequate context window for retrieved authority and answer-pattern instructions
- Demonstrated reliability at reasoning over provided context rather than answering from memory
- Support for parameter-efficient fine-tuning methods, such as low-rank adaptation, which reduces trainable parameters while preserving base weights (Hu et al. 2021), and quantized fine-tuning, which further reduces hardware burden (Dettmers et al. 2023)
- Calibration behavior
- Serving stability, with inference settings, system prompts, and model version recorded for later auditability

A model should not be selected merely because it is popular or newly released. The implementation must justify the choice in terms of domain sufficiency, reproducibility, and controlled deployment.

6.3 Baseline Evaluation and Staged Training

Before fine-tuning, the selected base model should be evaluated in its unmodified state on a representative subset of insurance prompts spanning all three UCF tiers and the edge-case categories from Section 5.4. It should be scored on the six UCF dimensions plus implementation diagnostics: refusal appropriateness, missing-fact recognition, context adherence, and source-hierarchy discipline. This baseline is necessary for two reasons. It shows whether fine-tuning improves the model rather than merely changing its style, and it identifies the failure modes the training data must address.

Training then proceeds through five stages, each evaluated separately.

Domain instruction tuning teaches insurance guidance structure, answer patterns, source hierarchy, and safe caveating through supervised examples, including controlled incompleteness. The model is rewarded for identifying a missing state or missing policy language rather than filling the gap with generic advice.

Retrieval-context tuning teaches the model to answer using structured retrieved context, respect source priority, and recognize insufficient context. Deterministic retrieval is only useful if the model treats it as controlling rather than optional background. This stage should also teach citation discipline.

Contrastive sufficiency tuning uses paired insufficient and sufficient responses to teach that a fluent answer can still fail. The term contrastive here refers specifically to curated paired data, an insufficient response and a sufficient response to the same prompt, used as training pairs for standard preference-alignment algorithms such as DPO or ORPO. It does not imply a specialized token-level contrastive loss function, consistent with this protocol's tool-agnostic posture. For the question in the worked example of Section 4.6, an insufficient response might read: "Yes, you generally have thirty days to appeal a homeowners denial." It is fluent and direct, but it states a specific deadline the system never verified against the user's state or policy. A sufficient response instead reads: "Appeal deadlines for a homeowners denial can be set by your policy, by state law, or both, and they aren't always the same number. Check your denial letter and policy for the exact deadline stated there, and don't rely solely on what the adjuster told you verbally. If you can't find a clear deadline, contact your state's department of insurance, which can also tell you how to file a complaint if you believe the denial was handled improperly." The second response is not more confident-sounding, but it is the one that protects the user's actual appeal rights.

Preference optimization then teaches the model which answer to prefer among several plausible ones, using domain-specific preference pairs. Recommended pairs include a grounded answer

over an ungrounded one, a caveated answer over an overconfident one, a clarifying question over a premature conclusion, a policy-dependent explanation over an unsupported coverage determination, a state-aware answer over a generic one, a procedural sequence over unordered advice, an escalation-aware answer over a self-contained one when risk is high, and a refusal or partial answer over unsafe compliance. Direct Preference Optimization is one applicable method, since it trains directly from preference data without a separate reward-model pipeline (Rafailov et al. 2023). This follows the general logic of aligning models to human-labeled preferences established in earlier instruction-tuning work, narrowed here to a domain-sufficiency objective rather than general helpfulness (Ouyang et al. 2022).

Finally, regression and safety testing checks the model against held-out prompts, edge cases, retrieval failures, and adversarial framings before any formal UCF validation.

6.4 Avoiding Memorization, and Safety/Refusal Tuning

A central risk in domain fine-tuning is that the model learns to recite current authority from training data. This is useful at first, but dangerous once law or procedure changes and memorized content keeps sounding authoritative. Training examples should therefore favor patterns that externalize current authority, such as "check the current policy language" or "this depends on your state," rather than reciting rules from memory. Evaluation should treat a state-specific answer produced correctly but without retrieved authority cautiously, since the architecture did not demonstrate proper grounding behavior even if the answer happened to be right.

Safety tuning in this protocol is domain-specific rather than generic. Refusal and escalation training should cover requests to exaggerate damages or mislead an insurer, requests for individualized legal strategy, requests to interpret policy language as a binding coverage determination without full document review, and situations involving releases or recorded statements given without understanding the consequences. The model should not over-refuse. A good refusal is a safe redirection, not a dead end, and the preference dataset should include cases where the model should answer directly. A well-calibrated model knows when to answer, when to ask, when to caveat, and when to escalate.

6.5 Reproducibility, Regression Testing, and Documentation

Whatever specific tools an implementation selects, it should document base model identity and version, parameter count, license terms, fine-tuning method, quantization method if any, adapter configuration, training hardware and data size, epochs, learning rate and scheduler, batch size, context length, and inference settings. This allows two teams fine-tuning the same base model with different configurations to have their outputs meaningfully compared. Retaining modular adapters, where feasible, supports versioning, rollback, and auditing, though each adapter configuration still requires its own testing under the UCF rather than being treated as a validation shortcut.

After training, the model should undergo internal regression testing across Tier 1 through 3 prompts and the full edge-case set from Section 5.4. Each result should be scored or coded for whether the model asked for missing facts when needed, avoided unsupported coverage determinations, identified deadlines without guessing, used retrieved context correctly, provided actionable next steps, and avoided both unsafe compliance and excessive refusal. Results should be compared against the pre-training baseline by dimension and failure mode. If accuracy improves but actionability declines, the model may have become too cautious. If safety improves but completeness declines, it may be refusing instead of guiding. If jurisdiction sensitivity improves but transparency declines, it may be asking for location without explaining why it matters. This suite should be versioned, and any later change to the model, adapter, retrieval scaffold, or answer-pattern library should trigger re-testing on the same suite.

The completed build should be accompanied by a model card documenting model name, base model and version, fine-tuning method, training-corpus summary and excluded source categories, domain scope and jurisdictions covered, retrieval-scaffold dependency, known limitations and professional-boundary rules, intended users and prohibited uses, evaluation status and regression-test results, UCF validation status, and the maintenance schedule and source-refresh protocol. The card should state explicitly whether the model has been internally tested, formally validated under the UCF, or is a research artifact only. It should also identify the SLM as a component of a system, the fine-tuned model, retrieval scaffold, source registry, classifier, and audit logging together, not the model weights alone, since none of the work in this section by itself establishes that the system reduces the observed sufficiency deficit. That claim requires the validation protocol in Section 7.

7. The Comparative Validation Protocol

7.1 Purpose and Validation Object

Sections 3 through 6 define how to select a feasible domain, stabilize the model with deterministic retrieval, engineer the content map, and fine-tune a localized SLM. None of that establishes that the resulting system reduces the sufficiency deficit. This section returns to the Universal Core Framework as the fixed measurement standard (Walcher 2026a) and specifies a narrow validation question: does a localized, retrieval-scaffolded, insurance-specific SLM produce more sufficient insurance-navigation guidance than general-purpose models under the UCF? A positive answer must be earned dimension by dimension and tier by tier. A model that improves jurisdictional caution while losing accuracy, or becomes safer only by refusing too often, has not solved the problem it was built to solve.

Before evaluation begins, the study should declare what is being validated. The SLM alone is the fine-tuned model without retrieval or classification, useful for ablation but not the deployment condition. The SLM system is the fine-tuned model operating inside the full retrieval scaffold, with audit logging; this is the primary validation object. The architecture is the general method of

combining a localized SLM, deterministic retrieval, and UCF validation, which a single insurance pilot can support only as a feasibility claim, not a cross-domain one. Extension to other domains requires separate feasibility analysis and validation.

7.2 Study Classification, Prompt Library, and Comparison Set

The validation study should be classified under the UCF's study-classification system before data collection (Walcher 2026a). A Pilot Application Study may be appropriate for early internal testing. It can show whether the system appears promising, but it should not support broad inferential claims. A Framework Validation Study may be appropriate if the purpose is to examine whether the UCF itself can evaluate a retrieval-scaffolded SLM system consistently. A Domain Extension Study is appropriate if the SLM is evaluated with a complete prompt set across all three tiers, multiple comparison models, and documentation sufficient to support limited comparative claims. A Full Implementation Study additionally requires a large, balanced prompt library, formal reliability analysis, complete reproducibility materials, and the inferential statistical analysis specified in Section 7.4.

For an initial execution with constrained evaluator time, expert-reviewer availability, or budget, this paper recommends treating the first cycle as a Pilot Application Study. This is a smaller, descriptive run that establishes whether the system appears promising and surfaces obvious failure modes, without claiming comparative superiority over general-purpose models. That is an honest and sufficient goal for a first cycle. It is also consistent with the UCF's own claim-governance rules, which require exactly this restraint whenever sample size or reliability falls short of Domain Extension requirements (Walcher 2026a). A claim that the SLM system materially improves insurance guidance over general-purpose models is a stronger claim. It requires escalating to at least a Domain Extension Study, and ideally a Full Implementation Study, once resources support the sample size specified in Section 7.4. Treating the fully powered comparative study as a second-phase deliverable, built on what the first cycle learns, keeps the protocol's claims tied to the evidence actually collected at each stage. The classification actually achieved should be declared early in the study's methods section, along with the claims it does and does not permit, and the UCF's claim-governance rules should apply without relaxation simply because the evaluated system was purpose-built for this pilot.

The prompt library should cover all three UCF complexity tiers plus the ten edge-case categories from Section 5.4. These should not be treated as trick questions, but as the conditions under which consumer harm is most likely. The library may use the insurance subset from the companion Reference Implementation (Walcher 2026b) as a starting point, but any prompt used in model development must be excluded from formal validation or clearly marked as non-held-out.

The comparison set should include, at minimum, the base model before fine-tuning, the fine-tuned SLM without deterministic RAG, the fine-tuned SLM with deterministic RAG, at least one frontier general-purpose LLM, and at least one general-purpose retrieval-augmented system where

available. This set separates several questions a single "SLM versus everything" comparison would collapse: whether fine-tuning improved the base model, whether retrieval materially improved the fine-tuned model, whether the localized system competes with broader high-capability models, and whether domain-specific scaffolding improves over ordinary retrieval-augmented answering. Model identity, version, access method, temperature, system instructions, retrieval status, and collection date should all be disclosed under the UCF's model-selection requirements (Walcher 2026a). The hypothesis should be framed narrowly: the localized SLM system is expected to show relative strength where localization and source hierarchy matter most, not superiority on every dimension.

7.3 Response Collection and Scoring

Each model should receive the same user prompt under standardized, disclosed conditions. Retrieval should be disabled for general-purpose LLMs unless the condition is explicitly a retrieval condition. The fine-tuned SLM without RAG should be tested on domain system instructions but without retrieved authority, to isolate what the weights alone contribute. The full SLM system should be processed through classification, filtering, retrieval, reranking, context assembly, and answer generation, with retrieved sources captured as part of the response record.

Responses should be scored independently on the UCF's six dimensions, using dual-track scoring where resources permit. Track A is human expert scoring by domain-qualified evaluators against the anchored rubric. Track B is automated or AI-assisted scoring, with the judge model, prompts, and calibration procedures fully disclosed. Human and automated scoring should not be assumed equivalent. Their concordance should be measured and reported by dimension, with lower agreement expected on the normative dimensions (safety, actionability, jurisdiction sensitivity) than on the descriptive ones (accuracy, completeness). This is consistent with the pattern the companion Reference Implementation is designed to test directly (Walcher 2026b). The instrument should also capture implementation-level diagnostic fields: did the response identify missing jurisdiction or policy language, did it identify deadlines or deadline uncertainty, did it avoid an unsupported coverage determination, did it use retrieved sources correctly. These fields explain UCF scores without replacing them.

The study should also code behavioral signatures under the UCF's framework (Walcher 2026a): overconfidence, actionability inflation, procedural drift, hallucinated institutional facts, uncertainty suppression, and persona overfitting, among others. For a localized SLM system, additional implementation-specific codes are useful: source overclaiming, retrieval neglect, policy substitution, jurisdictional shortcutting, and excessive safety refusal. These implementation-specific codes should be reported as exploratory unless preregistered. They remain descriptive; a coded behavioral signature is an observable output pattern, not a claim about the model's internal intent.

7.4 Statistical Analysis Plan

The statistical plan should be pre-registered before data collection and scaled to the study classification declared in Section 7.2. For a study intended to support comparative claims, whether Domain Extension or Full Implementation, the recommended design is a mixed-effects model with responses nested within prompts, and prompts nested within tier and edge-case category, following the form

Score = Model Condition + Tier + Framing + Edge-Case Category + (Model Condition × Tier) + (1 | Prompt),

estimated with standard mixed-model software (Bates et al. 2015). Model condition, tier, and framing are fixed effects, and prompt is a random effect, so that responses from multiple models to the same prompt are not treated as independent observations. Significant omnibus effects should be followed by pairwise comparisons: base model versus fine-tuned SLM, SLM without RAG versus SLM with RAG, and SLM system versus frontier LLM, with effect sizes reported as Cohen's d. Because this design tests multiple pairwise comparisons across six dimensions, the analysis should apply a false-discovery-rate correction across the family of comparisons rather than an uncorrected significance threshold. This avoids inflating the apparent number of dimensions on which the SLM system "wins" (Benjamini and Hochberg 1995).

Sample size should be set by an a priori power analysis rather than by convenience. As a benchmark, detecting a medium effect (Cohen's $f = 0.25$) in a two-way Model Condition × Tier design at 80% power and $\alpha = 0.05$ requires on the order of 60 to 65 observations per cell. With five comparison systems and three tiers, a fully balanced design of this kind implies at least 900 to 1,000 scored responses before dual-track reliability subsampling. This should be treated as a floor rather than a target. This benchmark applies specifically to the comparative-claim phase described in Section 7.2. A first-cycle Pilot Application Study is not expected to meet it and should scope its claims to descriptive and feasibility findings accordingly. A study falling meaningfully below this threshold should be classified and reported as a Pilot Application Study under the UCF (Walcher 2026a), and it should not present its comparative findings with inferential claim strength.

Reliability itself should be reported using Krippendorff's alpha for categorical and ordinal rater agreement (Krippendorff 2018), with a pre-registered target of $\alpha \geq 0.70$ per dimension, and using intraclass correlation coefficients for continuous composite-score agreement between human and automated tracks. Both statistics should be reported with confidence intervals, not as point estimates alone.

7.5 Ablation Testing and Primary Outcomes

Because the proposed system has multiple components, a positive result requires ablation testing to determine which component produced it. Recommended conditions are the base model alone, the base model with deterministic RAG, the fine-tuned SLM without RAG, the fine-tuned SLM with RAG, and, where feasible, the fine-tuned SLM with retrieval but without source-priority reranking

or without edge-case answer-pattern constraints. If the base model with RAG performs nearly as well as the fine-tuned SLM with RAG, the retrieval scaffold is likely doing most of the work. If removing source-priority reranking materially reduces performance, the deterministic scaffold is validated as more than ordinary retrieval.

The primary outcome should be defined as improvement, without material decline in accuracy or transparency, in the combined normative sufficiency profile: actionability, safety, and jurisdiction sensitivity, on Tier 2 and Tier 3 prompts specifically. These are the dimensions and tiers most directly tied to the rationale for localization. A model that improves only Tier 1 definitional accuracy has not addressed the deficit that motivated the build. Secondary outcomes include completeness, transparency, tier-specific and edge-case performance, behavioral-signature prevalence, human-automated concordance, and retrieval-failure handling. The study should avoid relying on a single composite score, since composites can hide the exact tradeoffs that determine whether a system is actually deployable.

7.6 Claim Governance, Failure Modes, and Revalidation

All claims following validation are bounded by the UCF's claim-governance rules (Walcher 2026a). The study should not claim that the model understands insurance, reasons like an expert, or eliminates the need for professional review. Permissible claims include that the localized SLM system outperformed the base model on specified UCF dimensions, or that the pilot supports further domain-extension testing. Prohibited or unsupported claims include that the system is safe for all insurance consumers, that it determines coverage, or that it resolves high-stakes consumer guidance across domains beyond the one evaluated.

The study should define validation failure modes in advance, since a negative result is still useful if it identifies which assumption was wrong. Potential failure modes include no improvement over the base model, improvement over the base model but not over frontier LLMs, improvement in accuracy paired with decline in safety, strong performance on clean prompts paired with failure on edge cases, retrieval context ignored or misused, retrieved authority overclaimed, and low human-rater or human-automated reliability. Each failure mode should map to a likely corrective action. Retrieval neglect suggests more retrieval-context tuning. Excessive refusal suggests additional preference pairs showing safe partial answers. Jurisdictional shortcutting suggests classifier improvement or stronger generation constraints.

Validation is not a one-time event. Revalidation should be triggered by a base-model or adapter update, a training-data or retrieval-corpus update, a source-registry or classifier change, a major statutory or regulatory change, a policy-form revision, or a recurring failure pattern detected in audit logs. Maintenance should be organized into tiers. Tier 1 is source refresh only, requiring targeted regression. Tier 2 is a source-hierarchy or answer-pattern change, requiring regression testing across affected task categories. Tier 3 is a model or training-data change, requiring broader validation including comparison to the prior version. Tier 4 is domain expansion or major regulatory change, requiring a new validation study.

As an illustrative default in the absence of domain-specific information, a quarterly source-refresh cycle paired with an annual full revalidation is a reasonable starting cadence for insurance. The actual cadence should be set once the update-detection layer in Section 4.8 has run for at least one full cycle and the domain's real rate of regulatory and policy-form change has been observed directly, rather than assumed in advance. The study's outputs should include the validated prompt library, raw model responses, the scoring dataset, human-rater calibration materials, the human-automated concordance report, behavioral-signature coding data, retrieval audit logs, ablation results, the failure-mode taxonomy, statistical analysis scripts, the model card, the revalidation schedule, and the claim-governance statement, so that deployment status follows the evidence rather than aspiration. The system should never be treated as validated in a form other than the one that was actually tested. A fine-tuned SLM answering without retrieval, audit logging, or refusal thresholds is not the system evaluated in this section, whatever the underlying weights.

8. The Dual-Role Deployment Paradigm

8.1 Two Roles, Two Risk Profiles

A validated localized SLM system has two distinct possible deployment roles, and a system validated for one is not automatically validated for the other. The first role is direct consumer guidance within a bounded domain, which exposes the user to immediate reliance risk. A flawed answer may cause a missed deadline, a misread policy provision, or an unwise settlement. The second role is evaluation and auditing of other AI-generated consumer guidance, which creates a different risk. A flawed scoring agent may systematically over- or under-rate third-party outputs and distort research findings rather than harming a user directly. Each role requires its own operational controls, evidence threshold, and claim boundary, and the validation target must match the deployment role actually intended.

8.2 Direct Consumer Guidance

In this role, the system responds directly to consumer questions within the validated domain: explaining policy concepts, helping organize a claim, identifying missing documents, and flagging when professional review is needed. It is the most hazardous role, since a user may not distinguish general guidance from individualized advice, may omit facts that change the answer, or may delay contacting a professional because the system sounded confident.

Deployment should therefore be scoped tightly to what was actually validated: domain, jurisdictional coverage, insurance line, task category, and model version. Controls should include domain-boundary detection, jurisdiction control, policy-dependency control, deadline control, professional-threshold control, retrieval grounding for any localized or rights-affecting claim, and full audit logging. These controls are part of the deployment condition, not optional interface features. Removing any of them changes the system away from the one that was validated.

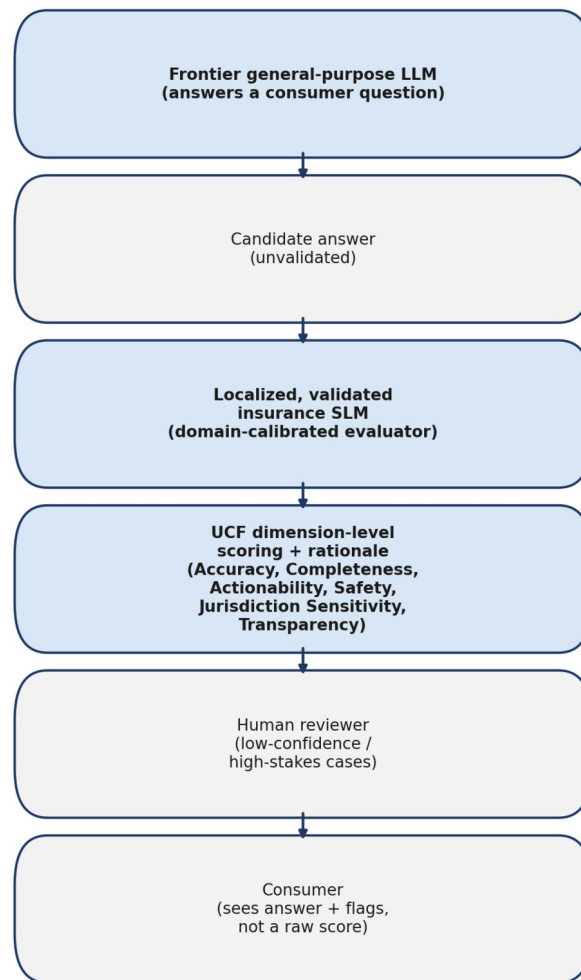
Consumer-facing output should generally contain five elements: a plain-language answer where one can be given safely; the key facts that could change the answer; a sequenced next-step list; a source-and-uncertainty statement disclosing whether the answer rests on general guidance, retrieved state material, or missing authority; and an escalation statement identifying when to contact the insurer, regulator, or a qualified professional. A generic "not a lawyer" disclaimer is not a substitute for this structure. The answer should name the specific reason for caution rather than gesture at caution in the abstract.

Recommended evidence thresholds, in increasing order of public exposure, are as follows. Research artifact only means the system is built and internally tested but has not completed formal UCF validation. Validated for controlled evaluation means UCF-aligned validation has occurred but public use is not yet approved. Validated for supervised consumer education means draft responses are subject to human review. Validated for bounded public guidance means the system may answer direct questions within a narrow validated scope with grounding, logging, and escalation controls. Validated for monitored deployment means the system operates publicly while audit logs, feedback, source updates, and regression tests are continuously monitored. A single successful comparative study may justify the earlier stages; it should not automatically justify the last one. The governing principle is that the system should not operate where foreseeable consequences exceed the validation evidence.

If deployed for direct consumer guidance, governance should include, at minimum: a visible scope statement; source-grounding disclosure; professional-boundary warnings; escalation pathways; a user feedback mechanism and incident reporting; audit-log retention; periodic source refresh and regression testing after updates; human review for high-risk categories; and a withdrawal or rollback procedure. High-risk categories warranting stronger caveats or referral include imminent deadlines, large losses, denial of essential benefits, threat of cancellation, settlement releases, suspected fraud allegations, vulnerable users, litigation threats, and claims involving injury or death. A deployed system should also disclose its institutional context where relevant. A system trained partly on carrier materials, for example, should disclose its source weighting.

8.3 Evaluation and Auditing Role

In this role, the system does not answer consumer questions directly. Instead, it scores or critiques the outputs of other AI systems within the insurance domain, supplementing broad commercial judge models with a domain-calibrated one. The motivation is that a general-purpose judge model may share the weaknesses of the system it is judging, rewarding fluent but incomplete answers, missing jurisdictional overreach, or underweighting policy dependence. A validated insurance SLM trained to recognize those specific failure modes may be a more domain-appropriate scoring agent, usable in a supervised pipeline where its outputs are checked by human evaluators rather than treated as final. Figure 8.1 sketches this loop end to end. A frontier model's answer is routed through the localized SLM for UCF-dimension scoring before either reaching the consumer directly or being escalated to a human reviewer.

Figure 8.1. The SLM-as-Evaluator Deployment Loop*Figure 8.1. The SLM-as-Evaluator Deployment Loop.*

The evaluator role may be the more practical near-term deployment path for a validated localized SLM, for several reasons. It reduces direct reliance risk, since the model's output is used by researchers or reviewers rather than by consumers making immediate decisions. It supports scalability, since a calibrated SLM judge can triage large response sets and reserve expert attention for ambiguous cases. It improves domain sensitivity, since a domain-calibrated judge may detect insurance-specific omissions that general-purpose judge models miss. It creates a feedback loop, since failures the scoring agent detects can inform new edge cases and retrieval rules.

This role still requires its own validation, distinct from validation as a guidance engine: rubric fidelity, consistency across models, resistance to stylistic bias, and calibration against human domain experts. Calibration should proceed by constructing a scoring corpus of model-generated insurance responses across all three UCF tiers, obtaining independent human expert scores, measuring human-SLM concordance by dimension (expected to be lower for the normative dimensions than the descriptive ones), testing for systematic bias toward length or confidence,

defining escalation rules for low-confidence or high-disagreement cases, and publishing the judge model, version, scoring prompt, and calibration results in full.

Risks specific to this role include:

- Judge overconfidence
- Rubric drift, where the judge substitutes its own learned preferences for the UCF rubric over time
- Stylistic bias
- Format exploitation or structural mimicry, where a response closely mirrors the Answer Pattern Library's structure from Section 5.3 while concealing an inaccurate or subtly hallucinated fact, which a judge keyed to structural cues may fail to catch precisely because the structure looks correct
- Source asymmetry
- Circular validation, where the same architecture both generates and judges answers, reinforcing its own assumptions rather than testing them independently
- False objectivity

These risks are mitigated through human calibration, independent validation sets, disclosure of judge limitations, and periodic recalibration. The role is governed by the same principle as direct deployment: the model's operational role must not exceed its validation evidence.

If deployed as an evaluator, governance should include, at minimum: the published scoring prompt and rubric mapping; the judge model version; a summary of the calibration dataset; human-concordance results; known scoring limitations and bias checks; escalation rules for low-confidence or high-disagreement cases; an audit trail for scores; periodic recalibration; and disclosure whenever a score is AI-assisted. The scoring agent should not be the sole evaluator for high-stakes claims unless separately validated for that purpose. In most research and governance settings it should function as an assistant to human review, with human reviewers retaining authority where the outcome matters. Its version-control and revalidation schedule should be tracked separately from the models it evaluates, so that observed changes reflect target-model drift rather than judge drift.

Where the same underlying SLM is used for both roles, the two should be separated by a firewall: different system prompts, different use of retrieved sources, different output structure, different failure handling, different validation metrics, and different deployment thresholds. The evaluator system may be acceptable for supervised research use before the consumer system is acceptable for direct public guidance. This separation prevents the system from silently shifting from giving guidance to judging guidance without a corresponding change in prompt and audit state.

8.4 Relationship to the UCF

The dual-role paradigm extends the UCF without modifying it. The UCF remains the evaluation framework. The localized SLM system is either an object evaluated under the UCF, a tool used

to assist UCF scoring, or both, and it does not replace the UCF's dimensions, scoring requirements, or claim-governance rules (Walcher 2026a). The SLM protocol introduces no new evaluation metrics of its own. Every claim in Section 7 is scored on the same six dimensions, the same scale, and the same tiers used to diagnose the sufficiency deficit in the first place. This is what keeps the comparison interpretable rather than self-referential. A validated domain-specific SLM judge could, in principle, make automated scoring in future UCF-based studies more domain-sensitive and less dependent on broad commercial judge models that share the failure modes of the systems they evaluate. This should be introduced as an optional, separately validated extension in domains where sufficient calibration evidence exists, not as a required or universal component of the UCF, which is deliberately domain- and model-agnostic.

8.5 What Would Count as Success

For direct consumer guidance, success means the system outperforms relevant comparison models on safety, actionability, jurisdiction sensitivity, and transparency without sacrificing accuracy or completeness, and handles missing facts, retrieval failure, and edge cases in a controlled way. For evaluator deployment, success means the judge produces dimension-level scores with acceptable concordance with domain-qualified human evaluators, detects known failure modes, and remains transparent about its own limitations. For architecture-level generalization, success means something more modest. The insurance pilot supports the feasibility of localized SLM development in one structurally favorable domain and justifies further domain-extension studies. It does not establish that the same method works everywhere. The only defensible claim, in any of these cases, is the one produced by a controlled build, a grounded retrieval scaffold, a documented content map, and UCF-based validation.

9. Negative Findings as a Valid Outcome

This protocol is built around a conditional premise: if the companion Reference Implementation confirms a sufficiency deficit in one or more high-stakes consumer domains, a corrective architecture is worth specifying in advance, so that the field is not left improvising one under pressure. It is equally important to state plainly what should happen if that premise does not hold.

If comparative evaluation under the UCF demonstrates that frontier general-purpose LLMs, deployed behind a well-engineered retrieval layer and system-level instructions, already satisfy the UCF's performance thresholds across high-stakes consumer domains, then the localized SLM architecture proposed here would not be justified for those domains. It is worth being precise about what that alternative requires. The engineering work in that scenario, building the retrieval layer and the system instructions that keep the general model grounded, would still be done once by developers before deployment. It would not require any consumer to phrase a question skillfully, any more than a consumer today needs to know how a search engine indexes pages. The comparison in Section 7 tests whether that developer-side engineering, applied to a frontier

model, closes the gap that motivates this paper, not whether individual consumers can be taught to prompt more carefully.

That outcome should not be read as a failure of this research program. It would be a successful result. It would eliminate an unnecessary layer of architectural complexity. It would redirect effort toward evaluation methodology and system-level instruction design rather than model-building. It would spare the field the cost of building, maintaining, and revalidating localized systems that a well-configured general model already renders unnecessary.

This paper's contribution is therefore not tied to the localized SLM winning the empirical comparison specified in Section 7. It is tied to the existence of a disciplined method for finding out, one that treats the question as open rather than settled in advance. A negative result, reached through the feasibility analysis, build protocol, and validation plan specified here, is exactly as publishable and exactly as useful to the field as a positive one, provided both are reached through the same evidentiary standard. The outcome this protocol is designed to prevent is neither a positive nor a negative result. It is a localized SLM built and deployed because it seemed intuitively safer, without ever being tested against the UCF at all. That failure mode is available regardless of which architecture ultimately performs better, and guarding against it is this paper's actual claim to usefulness.

Limitations of the Protocol

Several limitations of this protocol are worth stating directly rather than leaving for a reviewer to discover.

The feasibility matrix and pilot selection in Part I rest on an anticipated result: that the companion Reference Implementation will confirm a sufficiency deficit in at least one domain. If that anticipated result does not hold, or holds in a different domain than insurance, the specific engineering choices in Part II may need to be revisited even though the general method would still apply. This protocol is a plan made in advance of evidence, and it carries the ordinary risk of any such plan.

The single-domain pilot design, defended in Section 2.3 as a deliberate isolation of variables, is also a real constraint on what this paper can claim. A successful insurance pilot would establish feasibility in one structurally favorable domain. It would not establish that the same feasibility matrix, registry design, or fine-tuning strategy transfers cleanly to a domain with a different mix of ground-truth availability, localization variance, and temporal volatility. Generalization across domains is future work, not a claim of this paper.

Maintenance cost is a genuine ongoing burden, not a one-time engineering cost. The registry, update-detection layer, and revalidation cycle specified in Sections 4.8 and 7.6 require continuing institutional commitment. A team that builds the system but does not sustain the source-refresh and revalidation cadence would be operating a system that no longer matches the one that was

validated, and the protocol has no mechanism to force that commitment; it can only make the requirement explicit.

The fully powered comparative study specified in Section 7.4 is resource-intensive. The 900-to-1,000-response floor, dual-track human and automated scoring, and formal reliability analysis assume access to domain-qualified expert reviewers and sustained evaluator time. A team without that capacity is not thereby excused from the UCF's claim-governance rules; it is limited to the more modest claims available at the Pilot Application Study level, which this paper treats as an acceptable first-phase outcome rather than a shortcut around the standard.

Finally, this protocol depends on tools, model architectures, and fine-tuning methods that are likely to change before it is executed. Section 6 addresses this by specifying selection criteria rather than fixed tools, but that also means the protocol cannot guarantee that any specific technique named here, such as low-rank adaptation or direct preference optimization, will remain the best available method by the time of implementation. An implementation team should treat the cited methods as illustrative of the criteria, not as a mandated toolchain.

10. Conclusion

This paper has proposed a theoretical protocol for calibrating and validating localized small language models under the Universal Core Framework, on the assumption, not yet tested at the time of writing, that the companion Reference Implementation will confirm a sufficiency deficit in at least some domains of high-stakes consumer guidance (Walcher 2026b). Part I limited the first reference implementation to insurance navigation, selected because it combines codified authority, state-bounded variation, policy-dependent reasoning, and meaningful consumer consequence more cleanly than the other domains considered, and because a favorable result there would imply the least about other domains. Part II specified the required scaffold: an authoritative registry, deterministic retrieval, active authority requirements, update detection, retrieval-failure governance, and auditability. It also specified the Content Map needed to convert insurance materials into weighted corpus artifacts and held-out evaluation prompts, the build strategy for a domain-calibrated model, and a comparative validation protocol with an explicit statistical plan, before turning to the two deployment roles that could follow a successful validation and to what a negative result would mean.

The resulting claim is deliberately bounded in both directions. A localized SLM should not be presumed sufficient because it is smaller, open, specialized, or fine-tuned, any more than a general-purpose LLM should be presumed sufficient because it is larger or more fluent. Nor is a localized SLM presented as the only credible corrective architecture. A frontier model with stronger retrieval, an agentic workflow, a symbolic-hybrid system, or an architecture not yet developed may ultimately outperform the approach specified here, and the comparative protocol in Section 7 is designed to let that evidence speak rather than to presuppose the answer. In high-stakes consumer guidance, sufficiency is an empirical and methodological question, tested

through observable outputs, controlled prompts, disclosed context, independent scoring, and claim-governed interpretation. Under that standard, the localized SLM proposed here is best understood as a candidate corrective architecture, one whose promise lies not in replacing professional judgment, but in creating a bounded, auditable, domain-calibrated system whose behavior can be measured against the same framework that identified the deficit in the first place. If the Reference Implementation shows that general-purpose models are in fact sufficient once properly evaluated, this protocol becomes unnecessary, and Section 9 explains why that would still be a useful finding in its own right. If it confirms the anticipated deficit, this paper is intended to make the next step interpretable rather than speculative.

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